Ashraf Sir

CSE-317

Data Communication

18.6.26

⇒ Data: relevant raw facts

|

⇒ What is science?

তথ্য উপাত্তের ভিত্তিতে পরীক্ষা-নিরিক্ষা, পর্যবেক্ষণ ও পদ্ধতি প্রক্রিয়ার লব্ধ ~~[illegible]~~ জ্ঞানকে বিজ্ঞান বলে।

⇒ What is computer?

Computer is an electronic device or machine which is capable to take input from environment and process the input for desired output.

⇒ Data communicat^n না থাকলে store করা data কোনো কাজে আসবেনা

Data → Information —processed→ Knowledge → Expertise

processed

- collection
- store
- process
- dissemination

Data communication 2 limitat^n

⇒ Telecommunicat^n এ distance is how long not mentioned

⇒ Different ways of communicat^n we will read only electronics

Google Classroom code: skivsxc2

Book:

Data Communicat$^n$ and networking 4$^{th}$ edit$^n$ Forouzan

Basic requirement of Data communicat$^n$

1) Sender, 2) Receiver, 3) Communicat$^n$ path, 4) Protocol

5) Content/Message

⇒ What is computer network?

If more than 1 computer that can interchange data

⇒ What is computer topology?

How the computers and the terminals remain connected in network is called topology

⇒ Terminal: Not independent computer. Computer এর সাথে যুক্ত devices এ গুলো যেমন CC camera

⇒ Data flow types ?

Ans: 3 → Simplex, Half Duplex, Full Duplex. Write with example

⇒ Packet switching

Connectivity ? nature packet

Landphone ? কথা বলতে গেলে সবসময় call engaged থাকায়

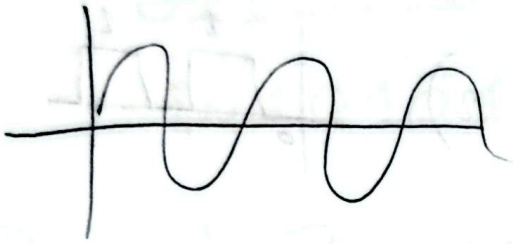
⇒ Circuit switching

Ashraf Sir

28.6.26

# CSE-317, Data Communication

## Data signal



⇒ Digital data signal

⇒ Analogue data signal

⇒ Signal ও কী বোঝায়? Ans: Data

## 2 limitation of Data communication

⇒ Data communication is mainly tele communication

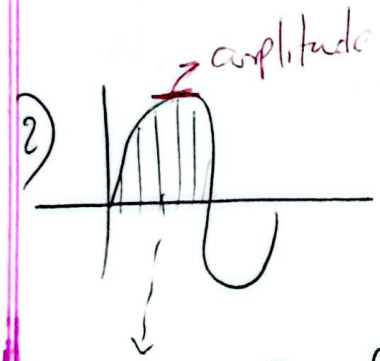
The distance is how long is not mentioned

⇒ There are many paths for data communication

We deal with electronic communication / electromagnetic " only

## Analog signal

1) Continuous

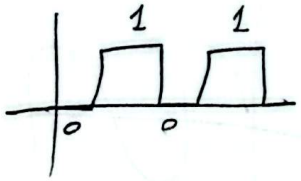
2) 


অনেকগুলো info থাকে

peak টা কত info দিচ্ছে calculate করা time এ

## Digital signal

1) Discrete

2) 


→ 0 এর meaning, specific

→ only one info

⇒ 0 can mean high/low

⇒ 0 " " any number as well

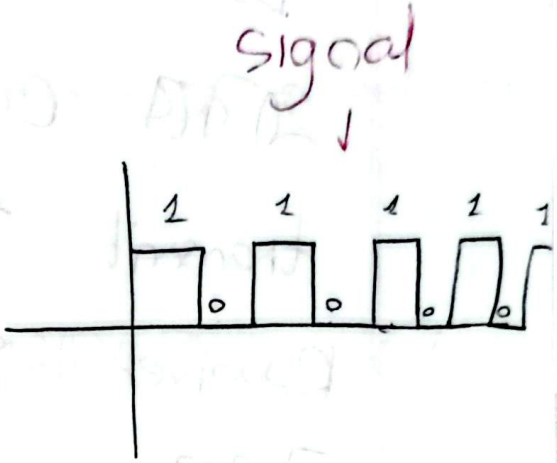
→ confined message equivalent to zero

## Periodic and non periodic

9 digit 의 periodic data

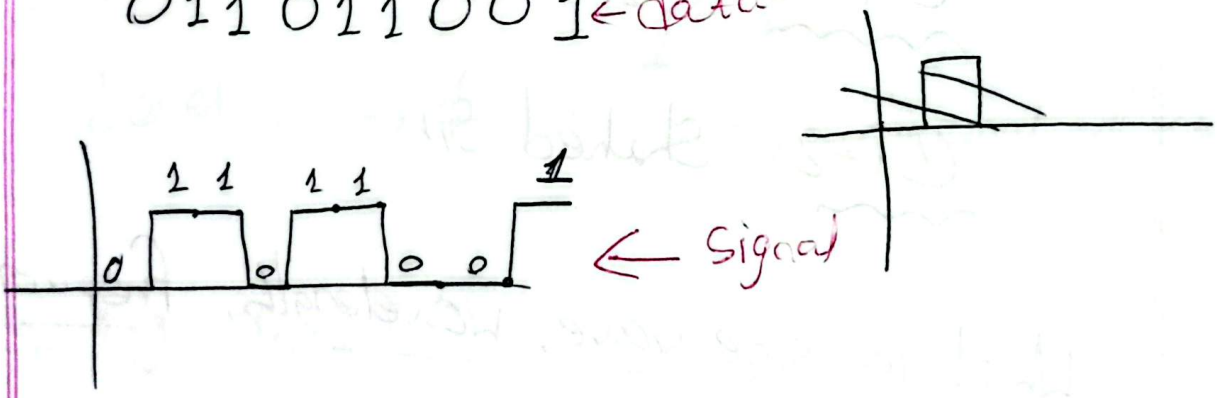
~~0 1 1 0 1 1 0 1 1~~ ← data

1 0 1 0 1 0 1 0 1




9 digit 의 non periodic data

0 1 1 0 1 1 0 0 1 ← data




## Sinusoidal signal

⇒ What is modulation and demodulation?

Data transmiss^n এর সময় data এর সাথে
আরও carrier then noise, এটা একসাথে
transmit হয়ে যাচ্ছে, Transmiss^n শেষে
Receiver প্রথমে receive করে demodulate
করবে

CT-1: 1st and 3rd week

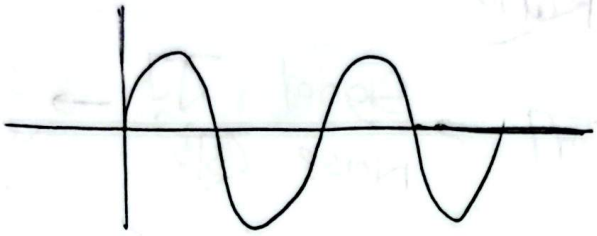
CT-2: Shahed Sir

What in sine wave, wavelength, frequency

frequency has relation to amplitude

Ashrat sir

CSE - 317 Data Comm 9.7.27



sine wave

CT-1 Ch-3

Example - 3.3, 3.4, 3.5, 3.6, 3.19 and 3.20

Fig - 3.4, 3.5

Data Analog, Digital signal analog, Digital periodic, non periodic

Essay: Sine wave and Data communication

important facts

6th week Thursday → Assignment

# Chp-1

⇒ Data, Data Communicatⁿ

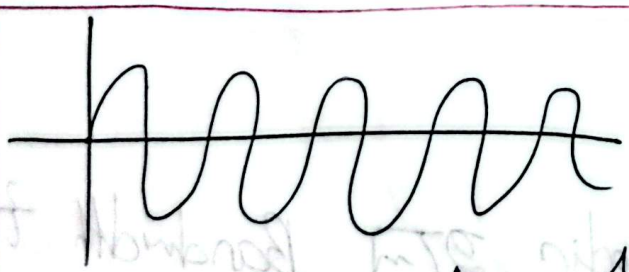
→ Why data communicatⁿ

→ Data flow with exmplᵏ

⇒ Network, Network Topology protocol

⇒ Justify data com course is very significant for the students of CSE

Chp-3 fig-3.4 Exam ~~Tote~~ Questⁿ



⇒ 12 periods in 1 sec

⇒ freq, $f = 12$ Hz

⇒ Time Period, $T = \frac{1}{f} = \frac{1}{12}$

What is frequency?

What is Time period?

Table-3.1 Units of period and frequency

fig-3.5

Fig-3.9

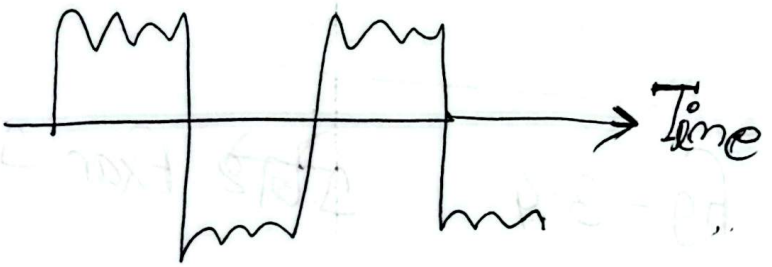



fig-3.11

⇒Non periodic 2Cn Bandwidth tn important

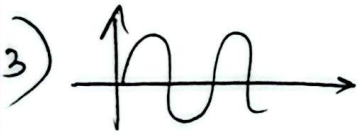
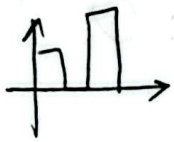
Exam-3.19

Exam-3.20

Ashraf Sir | CT-1

# Ashraf Sir CT-1

## Chp-3

| Analogue | Digital |
| --- | --- |
| 1) Continuous | 1) Discrete |
| 2) Infinite no. of values within a range | 2) Limited no. of values |
| 3)  | 3)  |
| 4) In data comperiodic analogue signal is used | 4) non periodic digital is used |

Ashraf Sir | Thursday missed class | 23.7.26

# Pulse modulation

Ashraf Sir 26.7.26

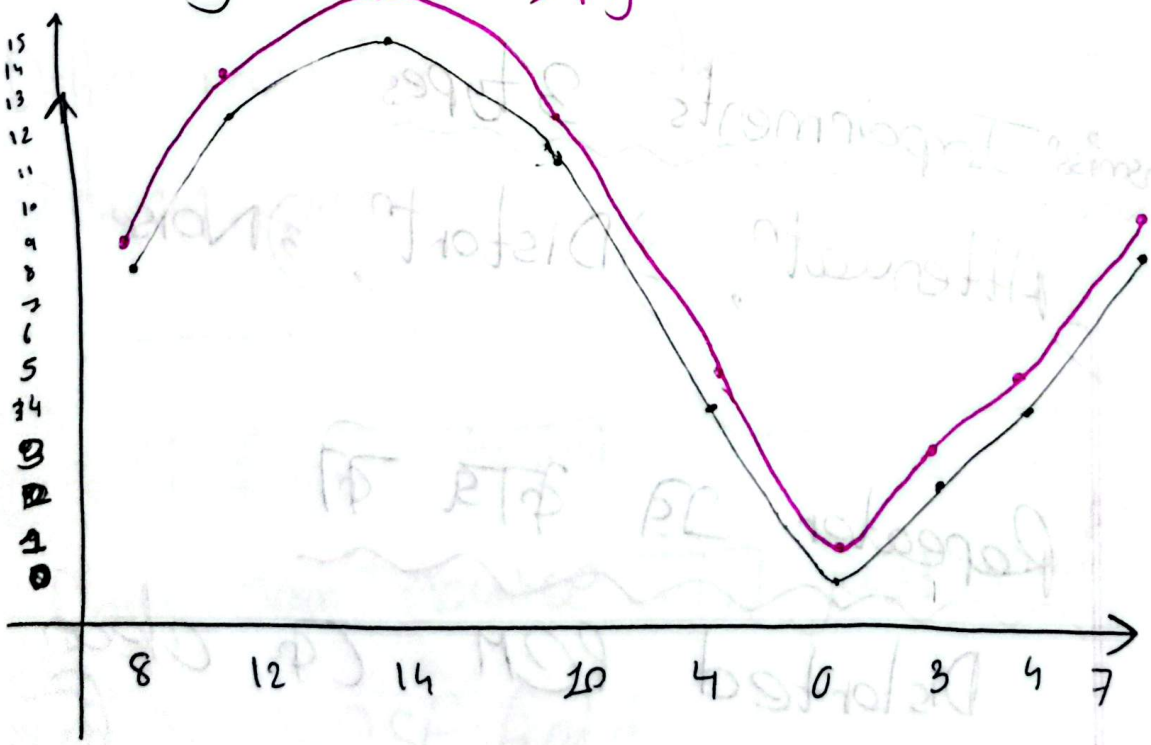
Encoding করব based on

8 12 14 10 4 0 3 4 7

⇒ 1 bit ~~left~~ ~~right~~ right shift করে waveform তৈরি করব

⇒ Number গুলোর waveform আগে আঁকব decimal এ Then arekta waveform আঁকব, 1 bit ~~left~~ ~~right~~ right shift করে 8→9, 12→13, Then convert to binary

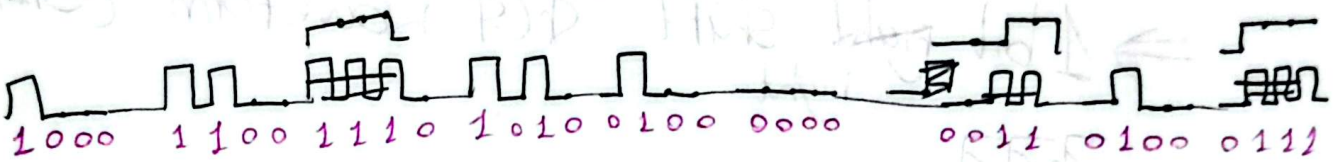
⇒ right shift করার পর waveform



5

8 12 14 10 4 0 3 4 7

1000 1100 1110 1010 0100 0000 0011 0100 0111



Right shift

9 13 15 11 5 1 4 5 8

1001 1101 1111 1011 0101 0001 0100 0101 1000



Transmiss" Impairments 3 types

1) Attenuatn, 2) Distortn, 3) Noise

Repeater का काम को

Distorted PCM को clean PCM में convert करना

## Advantages of repeater

Transmissn এর সময় original waveshape কে as it is রাখা

2 টা disadvantages আছে

Mid syllabus: ~~Chp~~

Sampler, Quantizer, Encoder

+ CT-1 syllabus

45 min

Delta modulatn এর definitn, advan, disadvan

30 min CT-1 এর আসার

Data com course টার justificatn কী in CSE field

8th week | Thursday

13.8.26

Ashpaf Sir | Chp-5

1) Modulat^n কাকে বলে?
2) Significants of every modulat^n
3) Types of modulat^n.
4) Impatance of carrier
5) Sinewave draw করতে হবে
6) Sinewave এর উপর base করে amplitude, frequency, phase
7) Analog to analog modulat^n কত প্রকার
8) Carrier কী?

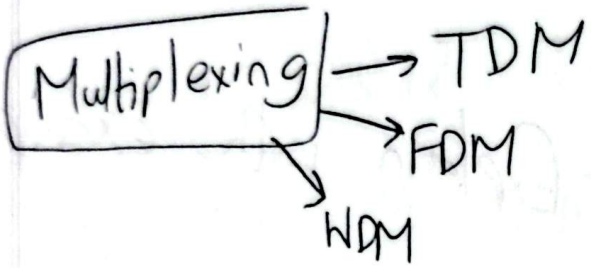
এই chp থেকে final এ 2Pmarks

Ashraf Sir

10th week
Thursday 27.8.26

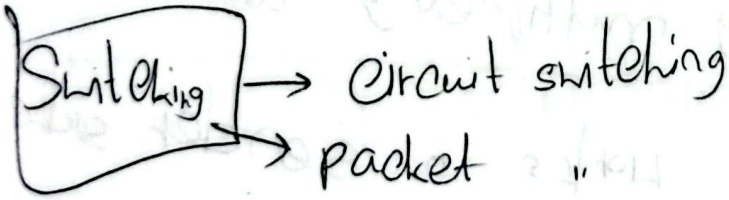
CSE - 317

Data Com




Multiplex: single input হয়ে multiple output
or, multiple " " single "

PBX → Private Branch Exchange




Circuit: path dedicated for a particular task

Packet: " not " " " " "

Exam

- Why we use Multiplexer?
- What is Multiplexer
- Why Multiplexing is used in Data Com

Multiplexing 3 types → 2 types in SecA
1 type in SecB

What is PBX
" " PABX
} Explain with example

Fig: 6.4 FDM process

Time sharing multiplexing advantages

Multiplexing works in sender side

De " " " receiver.

Why multiplexing is needed Given in slide

Multiplexing falls in Transport layer of OSI layer ??

Fig 6.5 FDM তে Maths আসবে

Ex-6.1

20-32 রেঞ্জে 12 টা পাব

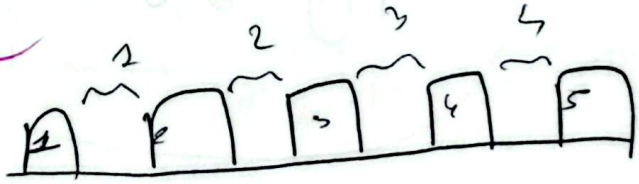
12 কে 3 ভাগে ভাগ করবে

each has 4




Ex-6.1 এ explanation টা imp

Ex-6.2 fig-6.7




Guard band

↳ Gap টা

⇒ 5 Chanels each 100 kHz

⇒ 4 Guard bands of 10 kHz

Minimum Bandwidth = 5x100 + 4x10 = 540 kHz

⇒ If Guardband is applied is Ex-6.1 then solve it

20 - 24  25 - 29  30 - 32

এরকম হবে তাহলে আগে থেকে হিট করতে হবে

⇒ If Guard is omitted in 6.2 then solve it

Fig 6.6 → This is must in Exam

## Mid syllabus

⇒ PCM → encoding quantizing sampling

⇒ Chp 1 + Chp 3

⇒ Topology

⇒ What is Top

⇒ Types of Top

CT-1 এর সবকিছু → Data কী, Components কী, protocol Data Com কী, Protocol বের করে Ques আসবে

Protocol এর standard কী কেন ব্যবহার হয়

⇒ Particular কোনো case এ কোন topology

⇒ Pulse code modulat block diagram

⇒ PCM How it works - Explanation

⇒ One bit right shift করে encoding করা

⇒ Data Com এ delta modulat used Why? Justify it's significance

⇒ Sine wave আঁকতে হবে → Frequency, Amplitude, Phase

12th week | CSF-317 | Ashraf Sir

Thursday 10.9.26

## Switching

Ques → Data com কীভাবে secured

CO3: Communicat engineering, security

CO3 থেকে quest আসবে

☆☆ switching কী, switching এর বিভিন্ন types

☆ Virtual circuit networks থেকে → 10 marks

Packet switching

আলাদা আলাদা path এ প্যাকেট তাই hacker সহজে track করতে পারেনা

☆☆ Comparison of switching types 8.9

☆ Datagram. draw a datagram network with 4 devices

13th week
Sunday

Ashraf Sir

13.9.26

⇒ Switching, why switching is important, types of switching, comparison of switching types.

⇒ Efficiency and delay in switching (fig-8.4), fig-8.6

⇒ Delay and efficiency in datagram networks (fig-8.7, 8.8)

⇒ Virtual ckt netwrks fig-8.10, 8.13, 8.19, 8.20, 8.21, 8.24, 8.25